SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I M.Dhanush Kumar /192525318 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Dhanush Kumar

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

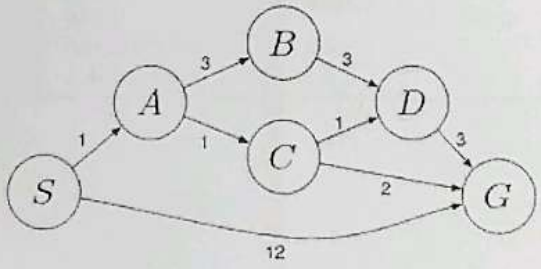
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

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CSA1710 - Artificial intelligence

CO1-AT1 - Analytical Problem solving

1) Water Jug Problem

→ Fill the 3-gallon jug completely.
→ Pour the 3-gallon jug into the 4-gallon
→ jug. The 4-gallon jug now contain 3 gallons.
→ Fill the 3-gallon jug again
→ Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full.
→ Empty the 4-gallon jug onto the ground.
→ Pour the remaining 1 gallon from the 3-gallon jug is full.
→ Fill the 3-gallon jug again and pour all 3 gallons into the 4-gallon jug.
→

The 4-gallon jug contains exactly 2 gallons.

2. ⓐ Mars Rover - Intelligent Agent

i. What are the percepts of this agent

- Images of the Martian surface.
- Soil and rock sample data.
- Temperature and atmospheric conditions.
- Signs of water on ~~this~~ other important materials.

ii. Characterize the operating environment.

- Partially observable: It cannot observe everything on mars at once.
- Partially deterministic: Actions may have uncertain outcomes because of terrain and ~~outer~~ environment
- Sequential: Each action affects future actions.
- Dynamic: Enviromnt conditions can change.
- Single

iii) How ~~can~~ can we evaluate the performance of agent?

- Number of useful samples collected and analysed
- Accuracy of scientific observations
- Area of Mars explored.
- Energy and time used.

iv. What agent architecture is most suitable? why?

- A model-based agent maintains information about its surroundings
- A goal-based selects actions that helps it achieve goals, such as reaching a location and Collecting samples.
- Combining both helps the rover plan routes. avoid obstacles and explores Mars efficiently.

3. Eight Queens Problem.

Problem formulation:

Component	Description
Initial State	An empty 8×8 chessboard
State	The positions of the queens already placed
Initial action	Place a queen in the first column.
Successor function	Place a queen in a square in next column
Goal Test	All 8 queens are placed without one another.
Path Cost	Usually one unit for each queen placed.

Lowest total Path Cost first.

Constraints:

Two queens must not share:

- The same row.
- The same column.
- The same diagnol.

one valid Solution

The row positions for columns 1 through 8 can be.

[1, 5, 8, 6, 3, 7, 2, 4]

This means the queen in column 1 is in row 1, the queen in column 2 is in row 5 and so on.

4) OLA Cab Booking - Problem - Solving Agent

A utility - based agent is suitable for the OLA Cab Cooking application.

Reason: The Customer Can Choose a Cab based on Price, Comfort, travel time and availability.

A Utility - based agent compares these factors and recommends a suitable option.

Explanation:

1. The Customer enters the pickup and destination locations.
2. The system finds available Cabs
3. it calculates the estimated fare and travel time for each Cap.
4. The Customer selects a suitable Cab.
5. The system Conflicts the booking and assigns a driver.

Result:

The Utility-based agent helps the Customer Select a Suitable Cab according to cost, Comfort and travel time.

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5. Uniform Cost search (UCS)

Problem: Find the least-cost path from starting warehouse S to destination warehouse G.

Algorithom:

Uniform Cost search expands the node with the lowest total Path Cost first.

steps:-

1. Start from warehouse S with cost 0.
2. List all neighbouring warehouse s and their route costs.
3. Select the ~~com~~ route with the smallest cumulative cost.
4. Expand that warehouse and calculate the costs of its neighbours

weighted graph

Route	Cost
S→A	1
S→G	12
A→B	3
A→C	1
B→D	3
C→D	1
C→G	2
D→G	3

UCS Calculation

	Total Cost
S→A→C→G	1+1+2 = 4
S→A→C→D→G	1+1+1+3 = 6
S→A→B→D→G	1+3+3+3 = 10
S→G	= 12

Result: The least-cost path is S→A→C→G, with a total cost of 4.